Arun B

Team Leader at Qubeats • Ph.D. Researcher (Applied Physics)

Qubeats • School of Physics, University of Hyderabad (CASEST Laboratory), India

arunskalikavu@gmail.com

+91 9496845082

Website

LinkedIn

Google Scholar

Hyderabad 500046, India

PROFESSIONAL EXPERIENCE & ENGINEERING LEADERSHIP

Team Leader

2026 February – Present

Qubeats

- Team leader of Magnetic Navigation application development for GPS-denied environments — driving core technology from prototype through to product development.
- Leading and coordinating an engineering team of 16 members across embedded systems, hardware instrumentation, sensor fusion, and navigation algorithms.

Electronics and Instrumentation Engineer

2025 Dec – 2026 February

Qubeats

- Engineered specialized electronics, sensor acquisition pipelines, signal conditioning circuits, and automated instrumentation test fixtures for magnetic navigation and sensing systems.

ACADEMIC & DOCTORAL RESEARCH APPOINTMENT

Senior Research Scholar (Ph.D. Candidate)

2019 – Present

School of Physics, University of Hyderabad (CASEST Laboratory)

Supervisor: Prof. K. C. James Raju

Status: Doctoral research completed; Ph.D. thesis submission in progress (Duration: 2019 – Present).

Dissertation Focus: Synthesis of soft magnetic FeGaB thin films, dynamic high-frequency magnetization dynamics, broadband VNA-FMR characterization, magnetoelectric voltage coefficients, and lightweight EMI shielding nanocomposites.

NATIONAL FELLOWSHIPS & ELIGIBILITY

CSIR-UGC National Eligibility Test (NET) – Junior Research Fellowship (JRF)

Qualified

Council of Scientific and Industrial Research, Physical Sciences

Graduate Aptitude Test in Engineering (GATE) — Qualified

Physics (PH)

State Eligibility Test (SET) — Qualified

Lectureship in Physical Sciences

Bachelor of Education (B.Ed) — Completed

Physical Science Education

SCHOLARLY IMPACT SUMMARY

Publications:	**22 Peer-Reviewed Journal & Conference Papers**
Citations:	**185+ Citations** (Google Scholar verified)
h-Index / i10:	**7 / 7**
Memberships:	**Life Member**, Magnetic Society of India (LM 904)

EXPERIMENTAL SETUPS & SCIENTIFIC HARDWARE ENGINEERED

1. Broadband Vector Network Analyzer Ferromagnetic Resonance (VNA-FMR) System

Designed CPW-based transmission fixtures and field sweeps from 0.1–20 GHz for Landé g-factor, effective magnetization, and intrinsic damping extraction.

2. Dynamic Magnetoelectric (α_{ME}) Voltage Measurement Bench

Integrated DC bias magnetic fields (up to 4000 Oe) with high-precision Helmholtz AC field coils and lock-in detection for multiferroic sensor testing.

3. Miniature In-Plane Electromagnet for RF Probe Station

Engineered an ultra-compact 800 Oe @ 1A electromagnet designed to fit directly onto high-frequency microwave probe station chucks.

4. Low-Cost Speed-Controllable Precision Optical Chopper

Developed an Arduino & BLDC drone motor based chopper with 3D printed slotted discs producing light modulation up to 2.8 kHz at <₹3,800 INR BOM.

PEER-REVIEWED PUBLICATIONS (CHRONOLOGICAL)

22. P. Anil, **B. Arun**, B. Saidulu, K. C. James Raju, V. V. Ravi Kanth Kumar, "Nitrogen-doped defect-induced dipolar polarisation in CNZF/NrGO/MWCNT nanocomposites for X & Ku-band microwave absorption", *Carbon Letters* 36(2), 907–935, 2026. doi:10.1007/s42823-026-01039-0

21. **B. Arun**, Jisna Rahman, T. S. Akhil Raman, Asokan Kandasami, N. E. Rajeevan, K. C. James Raju, "Swift heavy ion irradiation induced structural, morphological, and magnetic characteristics of FeGaB magnetic thin films", *Journal of Physics D: Applied Physics* 59(7), 075001, 2026. doi:10.1088/1361-6463/ae423d

20. Andrews Joseph, **B. Arun**, T. S. Akhil Raman, E. K. Suresh, K. C. James Raju, "Influence of microwave annealing on the structural, optical, and microwave dielectric properties of NBT thin films", *Indian Journal of Physics*, 2025. doi:10.1007/s12648-025-03845-0

19. Pasupuleti Anil, Andrews Joseph, **B. Arun**, K. C. James Raju, V. V. Ravi Kanth Kumar, "Enhanced X-band microwave absorption properties of cobalt ferrite/nitrogen-doped rGO nanocomposites", *Materials Science and Engineering: B* 323, 118859, 2026. doi:10.1016/j.mseb.2025.118859

18. Ateed Ahmad, **B. Arun**, Prabhu Rajagiri, K. C. James Raju, "Effects of light interstitial doping of Cu and Ag on the structural, magnetic and magneto-dynamics properties of Fe85Ga15 films", *Journal of Magnetism and Magnetic Materials* 615, 173556, 2025. doi:10.1016/j.jmmm.2025.173556

17. Santhoshkumar Dani, Arshad Ayub Hasamkal, Shivalingayya Gaddimath, **B. Arun**, K. C. James Raju, Ramcharan Meena, Pravin Kumar, Saif A. Khan, Khadke Udaykumar, "Ion Beam Irradiated Polyvinylidene Fluoride–Nickel Oxide Nanocomposites for Efficient Electromagnetic Shielding and Microwave Absorption", *Polymer Composites* 47(6), 5159–5174, 2026. doi:10.1002/pc.70481

16. Prabhu Rajagiri, **B. Arun**, S. Shanmukharao Samatham, Sankararao Yadam, Rajagopal Kurapati, Santhosh Kumar Adpa, Bunty Rani Roy, Gowrinaidu Babbadi, K. C. James Raju, "Uncovering the magnetism, universality class, magnetocaloric effect and ferromagnetic resonance of quenching-treated Co0.3Zn0.7Fe2O4", *Materials Chemistry and Physics* 345, 131176, 2025. doi:10.1016/j.matchemphys.2025.131176

15. Ateed Ahmad, **B. Arun**, J. Arout Chelvane, K. C. James Raju, "Structural, magnetic, and frequency domain ferromagnetic resonance studies of Fe82Ga18/Al2O3(001) films lightly doped with Cu and Ag", *Journal of Applied Physics* 137(15), 153901, 2025. doi:10.1063/5.0260810

14. S. Dani, A. A. Hasamkal, P. Patil, **B. Arun**, K. C. James Raju, K. Udaykumar, "Improved electromagnetic interference shielding in lightweight polyvinylidene fluoride–carbon nanotube composite films", *Journal of Applied Polymer Science* 142(11), e56589, 2025. doi:10.1002/app.56589

13. T. S. Akhil Raman, Andrews Joseph, **B. Arun**, E. K. Suresh, S. Chedurupalli, K. C. James Raju, "Structural and Dielectric Properties of Laser Crystallized BST Thin Films for Microwave Device Applications", *IEEE IWIS*, 135–137, 2023. doi:10.1109/IWIS61214.2023.10302775

12. E. K. Suresh, **B. Arun**, Andrews Joseph, T. S. Akhil Raman, P. Nikhil Mohan, C. Shivakumar, K. C. James Raju, "Role of buffer layers on the strain-induced insulator-metal transition of VO2 thin films: a review", *Critical Reviews in Solid State and Materials Sciences* 50(2), 137–160, 2025. doi:10.1080/10408436.2024.2400346

11. P. Nikhil Mohan, C. Thirmal, Andrews Joseph, **B. Arun**, K. S. Priya, P. Murugavel, G. Suresh, T. Vishwam, K. C. James Raju, "Microwave, ferroelectric and electromechanical studies of free standing blended electroactive polymer films", *Materials Research Bulletin* 181, 113105, 2025. doi:10.1016/j.materresbull.2024.113105

10. P. Nikhil Mohan, C. Thirmal, G. Suresh, **B. Arun**, Andrews Joseph, T. Vishwam, K. C. James Raju, "Microwave dielectric characterization of P(VDF-TrFE)-CuO nanocomposite free-standing flexible films", *Journal of*

Applied Polymer Science 142(6), e56462, 2025. doi:10.1002/app.56462

9. P. Nikhil Mohan, C. Thirmal, T. Vishwam, Andrews Joseph, **B. Arun**, K. C. James Raju, "Microwave dielectric characterization studies of PVDF-TrFE-Nafion blended films", *Ferroelectrics* 618(2), 474–485, 2024. doi:10.1080/00150193.2023.2273724

8. Shilpee Chauhan, **B. Arun**, K. C. James Raju, Muhamed Shafeeq M., D. P. Mondal, Avanish Kumar Srivastava, Pradip Kumar, "An ultrathin and flexible polymer composite film at a low fraction of reduced graphene oxide for Ku-band electromagnetic interference shielding", *Surfaces and Interfaces* 46, 103961, 2024. doi:10.1016/j.surfin.2024.103961

7. Manisha Thakur, Charanjeet Singh, K. C. James Raju, **B. Arun**, T. Tchouank Tekou Carol, A. K. Srivastava, "Fabrication of bicomponent Co–La substituted strontium hexaferrite for tunable microwave absorber application: Structural, morphological, reflection loss, input impedance metrics", *Physica B: Condensed Matter* 679, 415735, 2024. doi:10.1016/j.physb.2024.415735

6. T. S. Akhil Raman, **B. Arun**, C. Shivakumar, Andrews Joseph, K. C. James Raju, "Influence of oxygen pressure during deposition on the microwave dielectric tunability of Ba0.5Sr0.5TiO3 thin films in PLD process", *Applied Surface Science* 669, 160477, 2024. doi:10.1016/j.apsusc.2024.160477

5. **B. Arun**, Andrews Joseph, J. Arout Chelvane, K. C. James Raju, "Effect of RF power on magnetic properties and composition of FeGaB thin films", *AIP Conference Proceedings* 2995(1), 020177, 2024. doi:10.1063/5.0179425

4. S. A. V. Prasad, C. Srinivas, R. J. Kumar, **B. Arun**, K. C. James Raju, E. Ranjith Kumar, D. L. Sastry, "Thickness dependent microwave absorption in Mn-substituted Mg-Zn ferrites with quarter wavelength mechanism", *Materials Science in Semiconductor Processing* 214, 110881, 2026. doi:10.1016/j.mssp.2025.110881

3. E. K. Suresh, T. S. Akhil Raman, A. Othman, **B. Arun**, Andrews Joseph, K. C. James Raju, "Temperature stable (1 – x)BaV2O6-xBaZnV2O7 (x = 0.35–0.55) composite ceramics for low temperature co-fired ceramic applications", *Journal of Materials Science: Materials in Electronics* 34(14), 1159, 2023. doi:10.1007/s10854-023-10570-z

2. Ch. Srinivas, K. Naga Praveen, E. Ranjith Kumar, T. V. Chandrasekhar Rao, C. L. Prajapat, Sher Singh Meena, Pramod Bhatt, **B. Arun**, K. C. James Raju, D. L. Sastry, "Shielding performance of MnxNi0.8–xZn0.2Fe2O4 (0.1≤x≤0.7) for electromagnetic interference (EMI) in X-band frequency", *Ceramics International* 48(7), 9987–9997, 2022. doi:10.1016/j.ceramint.2021.12.205

1. Ch. Srinivas, K. Naga Praveen, E. Ranjith Kumar, Surendra Singh, Sher Singh Meena, Pramod Bhatt, T. V. Chandrasekhar Rao, Debashish Sarkar, **B. Arun**, K. C. James Raju, D. L. Sastry, "Microwave absorption properties of rare earth (RE) ions doped Mn–Ni–Zn nanoferrites (RE = Dy, Sm, Ce, Er) to shield electromagnetic interference (EMI) in X-band frequency", *Ceramics International* 48(22), 33891–33900, 2022. doi:10.1016/j.ceramint.2022.07.338

CONFERENCE PRESENTATIONS

ICMAGMA 2025 (IISc Bangalore) — Feb 2025

Poster: *"Effect of Swift Heavy Ion (SHI) Irradiation on Magnetic Properties of FeGaB Thin Film"*

Frontiers in Physics 2024 (Univ. of Hyderabad) Sep 2024

Poster: *"Development of VNA FMR and precise determination of Landé g-factor of FeGaB Thin films"*

ACMMT 2023 (Chiang Mai, Thailand) Nov 2023

Oral Presentation: *"Room Temperature Static Magnetic Properties of Pt/FeGaB/Pt Tri-Layer"*

ICACMP 2023 (GITAM Hyderabad) Oct 2023

Oral Presentation: *"Ferromagnetic Resonance Characterization of FeGaB Thin Films Using VNA"*

ICMAGMA 2023 (Hyderabad, India) Dec 2023

Poster: *"Frequency domain ferromagnetic measurements in FeGaB thin films"*

CORE COMPETENCIES & TECHNICAL EXPERTISE

Advanced Materials & Nanofabrication:	Nanofabrication, Cleanroom Operations, Semiconductor Device Fabrication, Thin Film Deposition, Sputter Deposition, Materials Characterization, Thin Film Characterization, Device Characterization, Ferromagnetic Resonance (FMR), XPS Analysis.
RF & Microwave Engineering:	RF Testing & Measurement, Microwave Characterization, On-Wafer RF Testing, Vector Network Analyzer (VNA), EMI/EMC Testing, Microwave Shielding Effectiveness Testing.
Embedded Systems & Instrumentation:	Embedded Systems Development, STM32 Development, Microcontroller Programming, Embedded C, Sensor Integration, Instrumentation Development, Test & Measurement Systems, NI LabVIEW, Hardware-Software Integration, Data Acquisition Systems (DAQ).
Communication Protocols:	SPI, I^2C, UART.
Design & Prototyping:	Autodesk Fusion 360, CAD Design, 3D Printing.
Tools, Software & Instruments:	Python, KLayout, OriginPro, STM32CubeIDE, VNA, Keithley SMU, Quantum Magnetometer (PPMS).
Leadership & Systems Engineering:	Technical Leadership, Cross-Functional Team Management, Project Planning & Execution, Resource Planning, Risk Management & Mitigation, Technical Mentoring, Systems Engineering, Research & Development (R&D).

Comprehensive Technical Index: Nanofabrication • Semiconductor Device Fabrication • Thin Film Deposition • Materials Characterization • Ferromagnetic Resonance (FMR) • XPS Analysis • RF & Microwave Characterization • On-Wafer RF Testing • Embedded Systems Development • STM32 Development • Sensor Integration • Instrumentation Development • Test & Measurement Systems • NI LabVIEW • KLayout • Python • OriginPro • Autodesk Fusion 360 • STM32CubeIDE • VNA • Keithley SMU • Quantum Magnetometer • SPI • I^2C • UART • Hardware-Software Integration • Systems Engineering • Technical Leadership • Cross-Functional Team

Management • Project Planning & Execution • Risk Mitigation • Technical Mentoring • Research & Development (R&D).

SPECIALIZED CLEANROOM & ANALYTICAL TRAININGS

- **INUP-i2i Hands-on Training on Nanofabrication & Characterization:** CeNSE, IISc Bangalore (Feb 2023).
- **INUP-i2i Nanofabrication Technologies Workshop:** IIT Bombay (Jan 2023).
- **ASSURE Workshop on Magnetism:** TIFR Hyderabad & Osmania University (Dec 2024).
- **Powder and GI-XRD Advanced Instrument Training:** School of Physics, UoH (Mar 2025).
- **Workshop on Ion/Laser Beams in Nano-Science (WINS-2025):** CFN, University of Hyderabad (Feb 2025).
- **Python for Data Analysis Hands-on Workshop:** SCIS, University of Hyderabad (Jan 2024).
- **Workshop on Rietveld Refinement:** MAMO College & SIAS Research Forum (Mar 2024).

MENTORSHIP & OUTREACH

M.Sc. Physics Project Mentorship

2024 – 2025

Mentored Suraj Chandra Joshi (23PHMP02), University of Hyderabad: *"Investigation of the magnetic properties of FeGaB thin films via VNA-FMR and high-frequency characterization"*.

TheSciencia.com & YouTube Channel

Active

Founder and primary educator delivering high-level CSIR-NET and SET Physical Sciences lectures across TheSciencia.com and YouTube.